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Journal of Accounting Research

DOI: 10.1111/1475-679X.12394
Journal of Accounting Research
Vol. 60 No. 1 March 2022
Printed in U.S.A.

CHICAGO BOOTH

How is Earnings News Transmitted to Stock Prices?

VINCENT GRÉGOIRE* AND CHARLES MARTINEAU†

Received 19 June 2020; accepted 25 May 2021

ABSTRACT

We examine the speed and mechanism of the price discovery process following earnings announcements in the after-hours market, a very illiquid trading environment. Prices reflect earnings surprises mostly through changes in quotes rather than through trades. Following positive announcement surprises, ask prices adjust quickly while bid prices are slower to adjust, and vice

*HEC Montréal; †Rotman School of Management and UTSC Management, University of Toronto

Accepted by Rodrigo Verdi. This paper is an extension of Martineau's Ph.D. thesis Chapter 1. We thank the anonymous associate editor and referee for valuable comments that have greatly improved this manuscript. Pat Akey, Daniel Andrei, Philippe d'Astous, Markus Baldauf, Oliver Boguth, Thomas Bourveau, Jonathan Brogaard, Mike Brolley, James Brugler, Murray Carlson, Tao Chen, Peter Christoffersen, Alex Edwards, Adlai Fisher, Mathieu Fournier, Will Gornall, Dale Griffin, Terry Hendershott, Alexandre Jeanneret, Ali Lazrak, Nan Li, Jiasun Li, Kai Li, Kin Lo, Russell Lundholm, Chay Ornthanalai, Andreas Park, Neil Pearson, Ryan Riordan, Kumar Venkataraman, Sheng Jun Xu, Bart Yueshen, Terry Zhang, Zhuo Zhong, Marius Zoican, and seminar participants at the UBC Finance and Accounting divisions, HEC Montréal, University of Virginia, Temple University, University of Colorado Boulder, Nanyang Technology University, University of Melbourne, University of Toronto Finance and Accounting divisions, Rice University, McGill University, Queen's University, York University, Tsinghua University, Binghamton SUNY, the NFA, and the SFS Calvacade Asia Pacific for their comments. We also thank Frank Hathaway for access to NASDAQ ITCH data, and Compute Canada for high-performance computing support. We acknowledge financial support from the NASDAQ OMX Educational Foundation, Montreal Exchange, Bank of Montreal Capital Group, Canadian Securities Institute Research Foundation, IVADO, and Social Sciences and Humanities Research Council of Canada (SSHRC). An online appendix to this paper can be downloaded at http://research.chicagobooth.edu/arc/journal-of-accounting-research/online-supplements

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versa for negative surprises. Returns computed from trade prices underestimate the speed and magnitude of price reactions following announcements relative to returns computed from quotes. These findings emphasize the importance of using quotes and not trade prices when examining intraday price discovery. Because firm announcements such as earnings generally occur in the after-hours market, using quotes is crucial as trading is sparse. We further illustrate the importance of quotes when examining the price discovery process around analyst recommendation revisions.

JEL codes: G10, G12, G14, M41

Keywords: after-hours market; analyst coverage; analyst recommendations; asymmetric reaction; disclosure; earnings announcements; liquidity; price discovery

1. Introduction

How do stock prices incorporate earnings news? Building on classical frameworks of price formation (Glosten and Milgrom [1985], Kyle [1985]), the model of Kim and Verrecchia [1994] predicts that stock prices reflect earnings news through the arrival of trades because liquidity providers are unsophisticated at processing such news. With the rise of automation and high-frequency traders becoming the dominant class of intermediaries, recent theoretical models (e.g., Hoffmann [2014], Budish, Cramton, and Shim [2015]) suggest that liquidity providers play a larger role in the price discovery of news through changes in quotes without the need for trades.

Whether liquidity providers or liquidity takers are now generally responsible for price discovery is key in identifying the appropriate methodology to measure the intraday price formation following news events. Recently, a growing number of studies in accounting use intraday data to examine the impact of corporate news on stock prices. Such recent studies include the analysis of high-frequency price formation around analyst conference calls (Matsumoto, Pronk, and Roelofsen [2011]), SEC insider trading filing releases (Rogers, Skinner, and Zechman [2016, 2017]), and analyst recommendations (Altınkılıç and Hansen [2009], Li et al. [2015]). As argued by Li et al. [2015], disentangling the effects of various news on prices using daily data is challenging when multiple events occur on the same day, thus the importance of using intraday data.

In this paper, we study how stock prices incorporate earnings news using high-frequency data in the after-hours market (4 p.m. to 9:30 a.m.) from 2011–2015 for S&P 1500 firms. Our objectives are to understand how quickly and through what mechanism, that is, through trading or changes in quotes, prices mostly incorporate earnings news. Further, we shed light on the price formation process in the after-hours market because 99.1% of earnings are announced in this environment, which is characterized by high trading costs, a high degree of information asymmetry, and low trading volume (Barclay and Hendershott [2003, 2004]).

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If today's price discovery generally occurs through changes in quotes and trading is sparse, using trade prices to compute returns can bias measurements of price discovery in multiple ways. First and foremost, this will cause trade returns to lag quote adjustments. Trade returns will then underestimate the speed and magnitude of price adjustments and more so when markets are illiquid. After-hours trading is indeed sparse; the median number of trades following earnings announcements in the after-hours market is only 16. Second, the use of trade returns can lead to a selection bias because it forces the exclusion of events with no trades even if price discovery takes place. For instance, we observe no trades following earnings announcements during the after-hours market for approximately 20% of our stock-earnings announcement observations.

We first examine the dynamics in cumulative returns computed from midquote and trade prices following earnings announcements in the after-hours market. We find midquote price adjustments to news occur quickly but are followed by price drifts for stocks with large surprises. We further show returns computed from trade prices significantly lag the adjustment in midquotes for S&P MidCap 400 and S&P SmallCap 600 stocks, while there is no significant difference for S&P 500 stocks. The fact trade prices lag midquote prices provides a first indication that price discovery is most likely attributable to liquidity providers updating prices through changes in quotes without necessarily relying on trades.

Midquote price drifts following announcements might indicate slow price discovery and profitable trading opportunities. Still, the ask and bid prices are what indicate to liquidity takers whether trading on the earnings surprise is profitable. We show bid–ask spreads are wide before announcements, in most cases already encompassing the post-announcement closing price at 4 p.m. of the following trading day. These large pre-announcement spreads indicate liquidity providers face adverse selection from other fast traders who can rapidly process the news.[1] Following announcements, the bid and ask prices gradually tighten around the post-announcement price, leaving no profits on average for informed liquidity takers that trade in the direction of the surprise.

Another key characteristics of spreads following announcements is the asymmetry in the speed of adjustment: following a positive surprise, liquidity providers instantly adjust the best ask price to the earnings surprise but only slowly adjust the best bid price, and vice versa for negative surprises. It is the slow adjustment in the bid price following positive news and in the ask price following negative news that generates midquote price drifts.[2]

[1] So and Wang [2014] show market makers demand higher expected returns prior to earnings announcements because of increased inventory risks. Wide spreads before news are consistent with the models of Hoffmann [2014], and Budish, Cramton, and Shim [2015].

[2] Li [2016] develops a trading strategy following earnings announcements in the after-hours market and also documents a slow price adjustment in midquotes. We show why midquotes slowly adjust to earnings news.

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Examining bid and ask price adjustments is thus important because simply considering adjustments in midquotes does not completely characterize the price discovery process. We find slow adjustments in the midquote do not indicate profitable trading opportunities on average in the after-hours market. A slow adjustment on one side of the quote is indicative of limited competition among liquidity providers or unresolved uncertainty. To our knowledge, we are the first to document that price discovery following earnings announcements occurs mostly through quote adjustments. These findings depart from the seminal model of Kim and Verrecchia [1994], in which liquidity providers are incapable of processing earnings news, and supports recent models of price discovery through immediate adjustment in quotes (e.g., Hoffmann [2014], Menkveld and Zoican [2017]).

We then quantify the magnitude and speed of price discovery for the best quote (i.e., the ask price following positive news and the bid price following negative news), midquote, and trade prices. Although the best quote instantly adjusts to earning surprises, both the adjustment in midquote and trade prices slowly approach the best quote. Because trading in S&P MidCap 400 and S&P SmallCap 600 stocks is sparse, the total response of trade prices to a one-standard deviation increase in earnings surprises one hour following the announcement is 15 and 32 basis points (or 17% and 35% in magnitude) smaller compared to midquote prices, respectively. The importance of stock-level liquidity in influencing price formation is significantly weakened when returns are computed using quotes instead of trades.

We further show liquidity providers charge a relatively higher trading price after negative news than positive news following announcements. Consistent with Johnson and So [2018], this suggests liquidity providers aim to minimize inventory risk in anticipation of news (e.g., conference calls). Moreover, computing returns using trade prices downplays this asymmetry. Over a 60-minute window following announcements, the response of trade returns to a one-standard deviation increase and decrease in earnings surprises lags best quote returns by 91 and 280 basis points, respectively.

We next examine the profitability of liquidity providers' executed orders to shed light on the implications of hidden liquidity—nondisplayed limit orders placed *within* the spread. A hidden order is only revealed to market participants when executed.[3] The use of hidden limit orders is much more common after hours than during regular trading hours. Trades against hidden orders may be profitable for liquidity-takers and thus contribute to price discovery. We find liquidity providers earn significant profits as large as 61 basis points per trade for hidden orders and 20 basis points for (nonstale) displayed orders, hence trades against hidden orders generally do not contribute to price discovery. This result further reinforces our findings that liquidity providers are sophisticated at processing public information.

[3] A hidden order must be placed within the spread to be executed because displayed quotes always have execution priority at the same price.

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The last empirical analysis examines price formation around analyst recommendation revisions, which are often triggered by firm news such as earnings announcements. We revisit an interesting result of Li et al. [2015], who find after-hours analyst recommendation revisions are followed by greater price adjustments than regular-hour revisions. We show this difference can be explained by the use of trade prices to compute announcement returns. Because trade returns are slow to incorporate news, they most likely do not fully reflect the information from the event that triggered the analyst revision by the time the revision is released. We find no substantial difference in the price adjustment between regular-hour and after-hours recommendation revisions when using midquote prices, further reinforcing the need to quantify price discovery using quotes and not trade prices.

Our findings suggest using quotes is important, especially when getting the timing of price discovery right is crucial, such as settings involving multiple news events within a short timeframe or when studying spillover effects of news events. Trade prices can lead to a selection bias, bias the price response to news and the price formation process, and veil the mechanism underlying price discovery. We detail our recommendations regarding the use of quotes (ask, bid, and midquote prices) to compute stock returns following after-hours news events in Section 6. We stress our results do not speak to price discovery studies using daily returns. Daily returns are computed from daily closing prices, which are determined by the daily closing auction.[4] The closing auction is usually the period of the day with the most considerable fraction of trade volume (Bogousslavsky and Muravyev [2020]).

Our paper contributes to a growing number of studies in accounting focusing on price adjustments to after-hours news (e.g., conference calls, analysts recommendations, and SEC disclosures) using high-frequency data (e.g., Matsumoto, Pronk, and Roelofsen [2011], Li et al. [2015], Rogers, Skinner, and Zechman [2016, 2017]). Our work is also related to studies examining intraday price formation following earnings news. One of the early studies is Patell and Wolfson [1984], who find profits of trading strategies based on forecast errors of earnings announced during regular trading hours dissipate within 10 minutes. Examining firms that began disclosing earnings overnight, Francis, Pagach, and Stephan [1992] find price discovery occurs during the following trading day and not overnight. More recently, Jiang, Likitapiwat, and McInish [2012] show a significant fraction of the total price variation over a trading day following earnings announcements for S&P 500 stocks now occurs overnight. In contrast to this last study, we focus on the price discovery process *within* the after-hours market. This allows us to investigate the mechanism and speed of price discovery directly. Our paper also relates to Santosh [2020], who examines the invariance hypothesis of Kyle and Obizhaeva [2016] following earnings announcements

[4] The auction price is determined by the clearing price where the supply and demand schedules cross.

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and interprets his result as evidence that price discovery occurs through trading, which he does not test directly.

2. *Theories of Price Formation Following Public News*

According to classical models of price formation, trades reveal investors' private information, and market makers (liquidity providers) adjust prices accordingly (e.g., Glosten and Milgrom [1985], Kyle [1985], Easley and O'Hara [1987]). Kim and Verrecchia [1994] build on these models to understand trade volume and liquidity dynamics around earnings announcements. The key assumption of their model is that earnings news provides information to a group of traders with better capacity to make judgments about a firm's performance. Such information is impounded into prices through trades as market markers are not sophisticated at processing earnings news. Panel A of figure 1 provides an illustrative example of the expected price path following positive earnings news for the quoted asking price and the transaction price. We assume one trade unit contributes equally to the incorporation of private information into the stock price and that all trades are buy market orders. When liquidity providers are uninformed, the figure shows the price path is a function of the occurrence of trades.

Today's financial markets have evolved into limit order books with competitive liquidity providers.[5] With the rise of automation, high-frequency traders (HFTs) have now become the dominant class of intermediaries. In contrast to Kim and Verrecchia [1994], recent models suggest liquidity providers invest in low-latency trading technology to avoid being "sniped" by faster traders. Speed technology gives liquidity providers the ability to quickly cancel (and revise) their quotes following news events, thus increasing the amount of price discovery that occurs through quotes (Hoffmann [2014], Menkveld and Zoican [2017], Budish, Cramton, and Shim [2015], Baldauf and Mollner [2021]).[6] Panel B of figure 1 illustrates the price discovery process when liquidity providers can process and adjust quotes immediately to reflect the positive news. In such cases, trades are not necessary for price discovery as informed trading is endogenous to the presence of an inefficient quote. Trade prices will only reflect the news when a trade occurs.

Understanding how prices adjust to earnings news is critical, not only to test theories of price formation but to guide empiricists on how to mea-

[5] Another class of models examine investors' choice between market and limit orders when endowed with private information, for example, Kaniel and Liu [2006], Goettler, Parlour, and Rajan [2009], Rosu [2020], and Brolley and Malinova [2021].

[6] In addition, because liquidity providers are unlikely to win every race to cancel their stale quotes, liquidity providers may also widen their posted spreads prior to news to compensate for the remaining pick-off risk. If stale quotes exist at the time of the announcement, Baldauf and Mollner's [2021] model predicts price discovery first occurs through trades by eliminating stale quotes, before reverting to updates in quotes.

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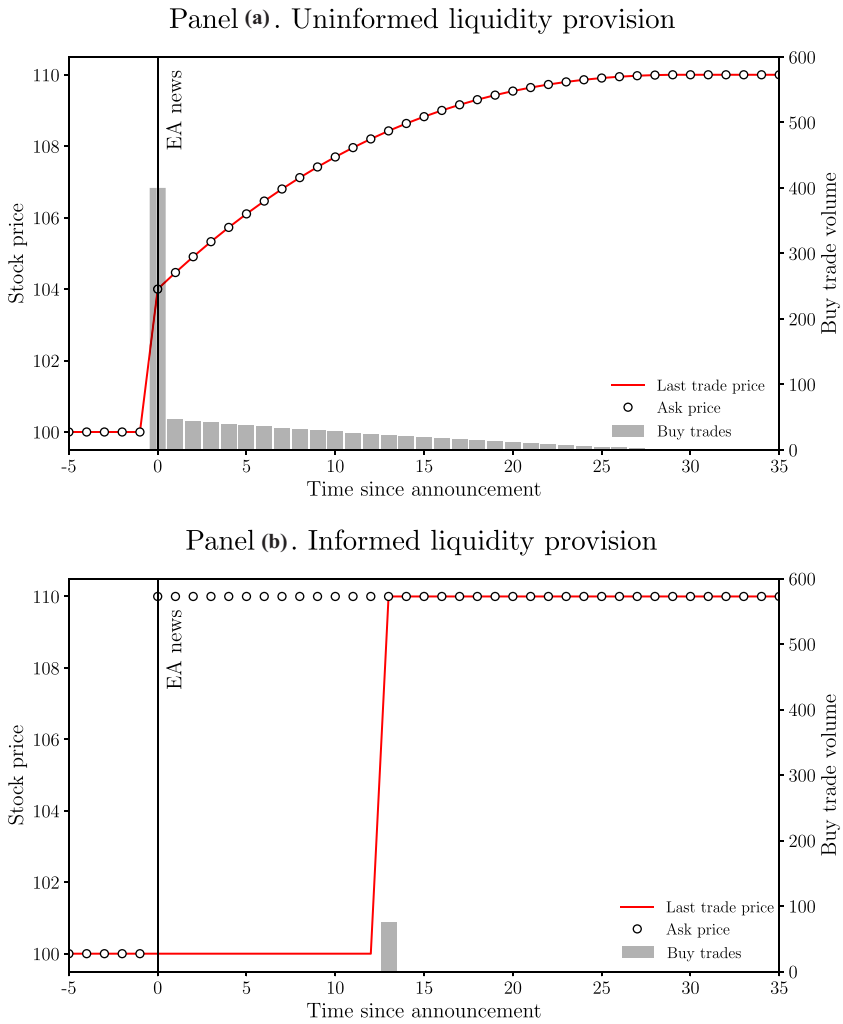

FIG 1.—An illustration of the expected price path when liquidity provision is uninformed and informed following earnings announcements. (A) Uninformed liquidity provision. (B) Informed liquidity provision.

This figure illustrates two examples of the expected path of transaction prices and quotes following positive earnings news when liquidity provision is uninformed (panel A) and informed (panel B) about earnings announcement (EA) news. The solid red lines correspond to the last transaction price, the circle markers correspond to the best asking price, and the gray bars correspond to trade volume. We assume the trade volume corresponds to buy market orders and that the true efficient price is 110. Under models with uninformed liquidity provision, price discovery only occurs through trades and ask prices track the trades. We further assume each trade (one unit of trade volume) contributes equally to the price discovery. Under models with informed liquidity providers, price discovery occurs immediately through quotes at the time of the announcement, and trades do not contribute to price discovery. In the extreme, trades do not occur as quotes are efficient and trade prices only reflect news when a trade occurs and have no price impact (e.g., noise trading).

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sure price adjustments. In settings such as Kim and Verrecchia [1994], there is no difference between computing returns using quotes or trade prices.[7] However, when liquidity providers are able to process news and adjust quickly quoted prices, it is critical for empiricists to calculate returns using quotes because returns computed from trade prices will lag the true price adjustment and thus bias downward the speed of price discovery (as shown in panel B of figure 1).

3. *Data Description*

The time coverage of this study is from January 1, 2011 to December 31, 2015. This period allows us to study price discovery around earning announcements, including the use of hidden limit orders by liquidity providers, which is prevalent in the after-hours market. This is not possible with more recent data due to a policy change by NASDAQ with respect to the information made publicly available about hidden orders. Although this information is crucial for our analysis, it has no bearing on our recommendation for researchers to use quote data when studying price discovery.

We use data from six different sources. Daily stock opening and closing prices are from the Center for Research in Security Prices (CRSP). Market capitalization and S&P index membership are from Compustat. Analyst forecasts and earnings announcements are from Thomson Reuters I/B/E/S. We obtain analyst revisions and more precise earnings announcement timestamps from RavenPack. Finally, high-frequency trade and quote data are from Nasdaq TotalView-ITCH and Thomson Reuters Tick History (TRTH, now Refinitiv Tick History) as supplied by the Securities Industry Research Centre of Asia-Pacific (SIRCA). Figure A1 of the Appendix summarizes the contribution of each data set to our study. We describe below the earnings announcement sample and the high-frequency data and present an overview of the trading environment in the after-hours market.

3.1 EARNINGS ANNOUNCEMENT SAMPLE

We first select CRSP stocks with share code 10 or 11 that belong to the S&P Composite 1500 index with NYSE, NASDAQ, or NYSE MKT as their primary listing. Each stock must have total assets and market capitalization data at the end of December of the previous calendar year in Compustat. The S&P 1500 combines three leading indexes—the S&P 500 (largest 500 stocks), the S&P MidCap 400, and the S&P SmallCap 600—to cover approximately 90% of the total U.S. equity market capitalization.

We then identify quarterly earnings announcements for our sample stocks using the dates and times recorded in the I/B/E/S database. Because I/B/E/S timestamps are not always accurate (see, Bradley et al.

[7] Computing returns using midquote remains the best because trade prices suffer from bid–ask bounce (Roll [1984]).

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[2014]), we use earnings news timestamps in RavenPack to improve the accuracy. We match 85% of the observations in I/B/E/S with timestamps in RavenPack. For the missing 15%, we rely on I/B/E/S timestamps.

Since the turn of the century, almost all firms now announce their earnings after hours (Michaely, Rubin, and Vedrashko [2013]). We focus only on after-hours earnings announcements (between 4 p.m. and 9:30 a.m. ET), which represent 99.1% of earnings announcements for S&P 1500 firms between 2011 and 2015.[8] We exclude weekend announcements (six observations) and all announcements between October 26, 2012, 4 p.m. and October 31, 2012, 9:30 a.m. because Hurricane Sandy forced stock markets to close for two trading days (111 observations). We lose 39 additional observations when merging with the TRTH data set described in the next subsection. Our final sample is composed of 25,265 firms-earnings announcements for 1,637 unique firms. The fraction of earnings announcements in our sample occurring on each day of the week, Monday to Friday, is 10, 23, 27, 34, and 6%, respectively.

We use analyst forecasts from I/B/E/S to compute earnings surprises:[9]

$$Surprise_i = \frac{EPS_i - E[EPS_i]}{P_i}, \tag{1}$$

where EPS_i is the earnings per share of earnings announcement i, and $E[EPS_i]$ is the expected earnings per share, proxied by the consensus analyst forecast. We scale the surprise using the stock price (P_i) five trading days before the announcement. We define the consensus forecast as the median of all analyst forecasts issued over the 90 days before the earnings announcement date. If an analyst revises their forecast during this interval, we use only the most recent forecast. Finally, we winsorize earnings surprises at the 1st and 99th percentile.

Table 1 presents descriptive statistics on sample firms, earnings announcements, and earnings surprises. The number of earnings announcements for firms in each S&P index (500, MidCap 400, SmallCap 600) is 8,964, 6,684, and 9,617, respectively. An S&P 500 firm has on average about three times more analyst forecasts (14 analysts) than an S&P 600 firm (5 analysts), suggesting more information production about these firms. The table further shows the inter-quartile range in earnings surprises for S&P 500, S&P 400, and S&P 600 stocks is 0.13, 0.17, and 0.30%, respectively, im-

[8] We do not address earnings news announced during regular trading hours because market conditions are vastly different than for announcements in the after-hours market. Moreover, some of these regular-hour announcements are the result of errors. For example, Google mistakenly announced its third-quarter earnings on October 18, 2012, at 12:30 p.m. instead of 4:30 p.m., which resulted in a three-hour trading halt.

[9] We choose earnings surprise over the simple random-walk approach because past research shows price drifts are more persistent following earnings announcements when surprises are calculated using analyst forecasts (see, Livnat and Mendenhall [2006]). Moreover, Walther [1997] finds sophisticated market participants put more weight on analyst forecasts than random walk forecasts.

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TABLE 1

Sample Stocks, Earnings Announcements, and Earnings Surprises Descriptive Statistics

	S&P 500	S&P MidCap 400	S&P SmallCap 600
No. observations	8,964	6,684	9,617
Market cap. (in million $)			
Mean	28,071	3,092	938
Median	12,637	2,730	789
Number of analysts			
Mean	14	9	5
Median	13	8	4
Earnings surprise (%)			
Mean	0.08	0.06	0.07
St. Dev.	0.28	0.35	0.49
P25	0.00	−0.02	−0.07
P50	0.04	0.04	0.06
P75	0.13	0.15	0.23
Earnings surprise distribution (%)			
Top quintile	15	18	26
Q4	21	20	19
Q3	26	22	13
Q2	25	20	15
Bottom quintile	13	19	27

This table reports the sample stocks, earnings announcements, and earnings surprises descriptive statistics by S&P index for S&P 1500 stocks. The S&P 1500 combines three leading indexes: the S&P 500, the S&P MidCap 400, and the S&P SmallCap 600. Earnings surprise quintiles are defined over the entire sample of earnings announcements. The sample period is January 1, 2011 to December 31, 2015.

plying small firms are more likely to have large surprises (in magnitude) than large firms. In line with this intuition, the table also shows S&P 600 stocks are almost twice as likely as S&P 500 stocks to have earnings surprises that belong in the top and bottom earnings surprise quintiles.

3.2 HIGH-FREQUENCY STOCK PRICES AND TRADING ACTIVITY DATA

We use two high-frequency databases that contain stock prices and trading activity data. The first is the TRTH data set that includes all trades and top-of-book quotes for U.S. exchanges and trades from alternative platforms such as dark pools, with millisecond timestamps. Although TRTH is a more comprehensive database than the NYSE Trade and Quote database (TAQ), the TRTH data we use for this paper are comparable to what is available in TAQ.[10] We match TRTH data with the earnings announcement sample on CUSIP for an overall match rate of 99.9%. Over the 2011–2015 period, our sample stocks have 19 billion trades for a total market value of $161 trillion.

We complement the TRTH data with order-level data from NASDAQ TotalView-ITCH.[11] The ITCH data contain all messages distributed

[10] See section IA1 of the online appendix for more details on TRTH data, the filtering we have done, and information for researchers who wish to apply these filters to TAQ data.

[11] Other studies that use NASDAQ ITCH include Hasbrouck and Saar [2013], O'Hara, Yao, and Ye [2014], and Comerton-Forde, Grégoire, and Zhong [2019].

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through NASDAQ's high-frequency feed, including messages that describe orders added to, removed from, and executed on the NASDAQ exchange for NASDAQ-, NYSE-, NYSE MKT-, NYSE ARCA-, and BATS-listed securities. We construct a message-by-message limit order book, where the book is updated whenever a new message enters the NASDAQ exchange.[12]

Trades are not signed in TRTH (nor in TAQ); the researcher must infer if a trade is buyer- or seller-initiated using a trade classification algorithm.[13] The main advantage of ITCH is that trades are accurately signed and linked to the liquidity-providing limit order. The only exception is for trades against hidden (i.e., nondisplayed) orders, for which we do not observe the corresponding limit order. Hidden orders are essentially limit orders that are not visible to other traders and, as we show, are widely used by sophisticated liquidity providers in the after-hours market. Hidden orders pose significant limitations to trade classification algorithms, and the use of order-level data from ITCH avoids these issues. Unfortunately, trade direction for trades against hidden orders are only available until July 13, 2014. See section IA1.3 of the online appendix for more institutional details regarding hidden orders.

After constructing the limit order book, we have for each trade the price, quantity, whether the trade was buyer- or seller-initiated, and information on the executed limit order such as the time it was posted to the order book or if it is hidden.

3.3 THE AFTER-HOURS MARKET LANDSCAPE

Liquidity in the after-hours market is very different from what market participants observe during regular trading hours. As shown in Barclay and Hendershott [2003, 2004], the after-hours market is much less liquid. Moreover, no liquidity providers have an obligation to participate during the after-hours market.

Figure 2 presents an overview of the trading landscape during regular- and after-hours trading for situations when earnings announcements do and do not occur. Panel A shows after-hours trading (pre– and post–regular-hour trading session) accounts for only 3% of the total daily dollar trading volume. Of that figure, 83.3% of trading occurs after the market closes (4 p.m. to 8 p.m.) and 16.6% occurs before the market opens (4 a.m. to 9:30 a.m.). Panels B and C show after-hours trading accounts for

[12] Compared to TRTH or the TAQ data set, NASDAQ ITCH data do not suffer from liquidity measurement problems and errors in trade-quote matching as reported in Holden and Jacobsen [2014]. However, processing these data and constructing the limit order book are computationally costly.

[13] Trade classification algorithms are not flawless (see, Chakrabarty, Pascual, and Shkilko [2015]). Because liquidity is mostly hidden in the after-hours market, it imposes significant limitations on the effectiveness of these algorithms.

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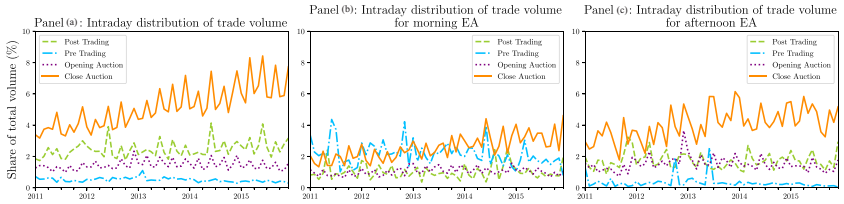

FIG 2.—Trade volume and fragmentation in the after-hours market. (A) Intraday distribution of trade volume. (B) Intraday distribution of trade volume for morning EA. (C) Intraday distribution of trade volume for afternoon EA.
Panel A of this figure shows the monthly share of total aggregate traded dollar volume during extended trading hours before the market opens (blue dashed-dotted line), after the market closes (green dashed line), at the opening auction (purple dotted line), and at the closing auction (full orange line). Panels B and C show the monthly shares of total aggregate traded dollar volume around earnings announcement (EA) windows for morning and afternoon announcements, respectively. We define the earnings window starting with the first trade following the closing auction before the announcement and ending with the closing auction of the trading session that follows the announcement. The sample period is January 1, 2011 to December 31, 2015.

approximately 2% and 4% of the total daily trading volume for morning and afternoon earnings announcements, respectively.[14]

Table 2 presents descriptive statistics of trading during regular market hours, in the after-hours market, regular hours before and after earnings announcements, and in the after-hours market following earnings announcements.[15] We highlight two important insights from table 2. First, trading in the after-hours market is minimal, even when an earnings announcement occurs. The median number of trades in the after-hours market with (without) an earnings announcement is 70 (17), 10 (7), and 7 (5) for S&P 500, S&P MidCap 400, and S&P SmallCap 600 stocks, respectively. Also, we find no trades in the after-hours market following earnings announcements for approximately 20% of events (untabulated). Of those events, 16%, 34%, and 50% are stocks that belong to the S&P 500, S&P 400, and S&P 600, respectively. We further observe a larger number of trades during regular hours around earnings announcements than regular hours outside the earnings announcement window. The second main insight from table 2 is that trades are larger (in shares and in dollar terms) in the after-hours market. Barclay and Hendershott [2004] show adverse selection risk is higher in the after-hours market and retail traders are being discouraged from trading. Therefore, traders that do choose to participate are more likely to be informed and sophisticated.

[14] Market fragmentation during regular trading hours and after hours also varies greatly. See section IA2 of the online appendix for more details.

[15] To capture the full view of market trading, the sample period for reported statistics starts on December 9, 2013, when the consolidated tape began reporting odd-lots (trades of less than 100 shares).

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TABLE 2

Regular- and After-Hours Trading Descriptive Statistics

	S&P 500					S&P MidCap 400					S&P SmallCap 600				
	RH	AH	RH Before EA	AH EA	RH After EA	RH	AH	RH Before EA	AH EA	RH After EA	RH	AH	RH Before EA	AH EA	RH After EA
Number of trades															
25th percentile	10,125	11	13,697	12	20,404	3,180	5	4,232	3	6,749	1,077	3	1,360	3	2,446
Median	16,900	17	21,826	70	35,432	5,123	7	7,242	10	12,197	1,827	5	2,395	7	4,692
75th percentile	28,494	31	36,705	623	62,106	8,675	10	12,685	59	24,023	3,205	7	4,465	27	9,795
Trade size in shares (%)															
<100	28	36	24	27	21	30	51	26	28	20	33	53	30	28	23
100–500	69	38	73	61	75	68	26	72	56	77	66	25	68	55	75
500–1,000	2	8	2	6	2	1	7	1	8	2	1	7	1	8	2
>1,000	1	17	1	6	1	1	16	1	8	1	1	14	1	9	1
Trade size in $ (%)															
<1,000	10	14	8	7	7	14	27	12	13	9	25	36	21	17	17
1,000–5,000	35	24	36	31	35	50	29	52	41	52	58	29	58	43	59
5,000–50,000	53	44	54	56	56	36	30	36	41	38	17	26	20	36	24
>50,000	2	19	2	6	3	0	14	1	5	1	0	10	0	5	0

This table presents statistics of trading activity during market regular hours (RH), after hours (AH), regular hours before and after earnings announcements (EA), and after hours following an earnings announcement by S&P index for S&P 1500 stocks. The S&P 1500 combines three leading indexes: the S&P 500, the S&P MidCap 400, and the S&P SmallCap 600. The sample period is December 9, 2013 to December 31, 2015 when odd-lots (trades with less than 100 shares) are reported to the consolidated tape.

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4. *Intraday Price Discovery Following Earnings Announcements*

We begin the analysis of earnings announcements by examining intraday price changes in the after-hours market to determine the speed, dynamics, and mechanisms of price discovery. We first examine price drifts around announcements using trade and midquote returns. We then study the dynamics in bid and ask prices around announcements and the role of liquidity in price discovery followed, by an analysis of the profitability of liquidity providers' trades.

4.1 PRICE DRIFTS IN THE AFTER-HOURS MARKET

Does price discovery of earnings news generally occur through changes in quotes or through the arrival of trades? To begin answering this question, we first graphically depict price adjustments to earnings surprises in the after-hours market following earnings announcements. We plot in figure 3 the cumulative returns using one-minute midquote (halfway point between the bid and ask prices) and trade prices around earnings announcements 5 minutes before to 60 minutes after the announcement by earnings surprise quintiles for the full sample and by S&P index in panels A–D.[16] Because bid–ask spreads can be very large, with the bid price sometimes equal to \$0.01 and the ask price equal to \$199,999, or simply nonexistent, we exclude observations for which the absolute difference between the ask and the bid divided by the midquote is greater than 20%. As is common in the literature, cumulative trade returns at time t are computed using the most recent trade up to time t, even if that trade occurred many minutes before t. For time intervals with no trades, we impose a return of 0%.

Figure 4 shows a clear demarcation in cumulative returns between earnings surprise quintiles. More importantly, panel A (full sample) shows the adjustment occurs more quickly for midquote returns than trade returns. For S&P 500 stocks (panel B), we find little difference in the adjustment between midquote and trade returns.However, the difference can be more than 1% for S&P 400 and 600 stocks (panels C and D), that is, less liquid stocks. This indicates price discovery is most likely attributable to liquidity providers updating prices through changes in quotes without necessarily relying on trades. Also, using trade returns rather than midquote returns biases downward the magnitude of the price response and the speed of adjustment to news. Therefore, an analysis of price adjustment around news announcements should be conducted using midquote returns and not trade returns. Figure 4 further shows little to no adjustment remains in the midquote 60 minutes after the news. Figure IA2 in the online appendix presents the cumulative returns during regular trading hours

[16] We choose 60 minutes as the endpoint because we want to limit the number of earnings announcements for which the buy-and-hold returns extend beyond 9:30 a.m. For 914 (3.6%) earnings announcements in our sample, the announcement is between 8:30 a.m. and 9:30 a.m.

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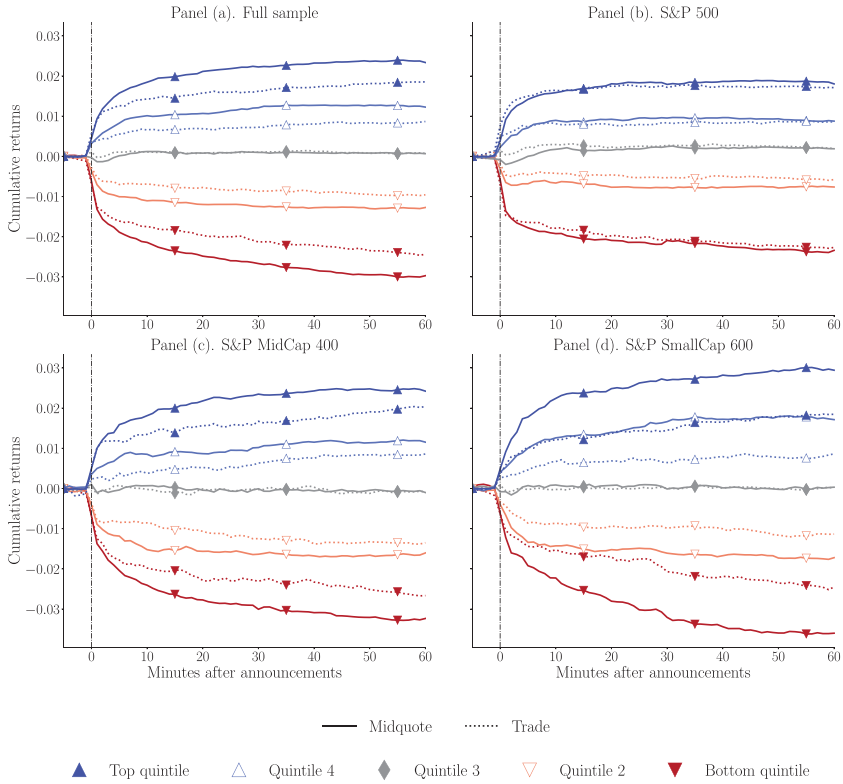

FIG 3.—Cumulative returns following earnings announcements in the after-hours market. (A) Full sample. (B) S&P 500. (C) S&P MidCap 400. (D) S&P SmallCap 600.
This figure shows the buy-and-hold returns (BHR) in the after-hours market from 5 minutes before to 60 minutes after the earnings announcement by earnings surprise quintiles. The BHR values for the full sample, S&P 500, S&P MidCap 400, and S&P SmallCap 600 stocks are presented in panels A–D, respectively. We define BHR for stock-earnings announcement i from time τ to T, $BHR[\tau, T]_i = \sum_{t=\tau}^{T} r_{i,t}$, where $r_{i,t}$ is the stock log return for the one-minute interval t. Midquote returns are computed using the midpoint between the best displayed bid and best ask price (solid line), and trade returns (dotted line) are computed using transaction prices. Trade returns are calculated from trade prices; for any minute interval t with no trades, we impose a return of 0%. The sample period is January 1, 2011 to December 31, 2015.

before and after earnings announcements and indeed shows most, if not all, price discovery occurs overnight, as cumulative returns during regular trading hours following the opening of markets are mostly flat.

Though our analysis focuses solely on S&P 1500 stocks, we show our results are also generalizable to the smallest stocks. We reproduce figure 3 for microcap stocks, that is, stocks smaller than the NYSE 20th size percentile for a total of approximately 1,800 stocks, and present the results in figure IA3 of the online appendix. We find comparable price adjustments to S&P 600 stocks, but midquotes are slower to adjust.

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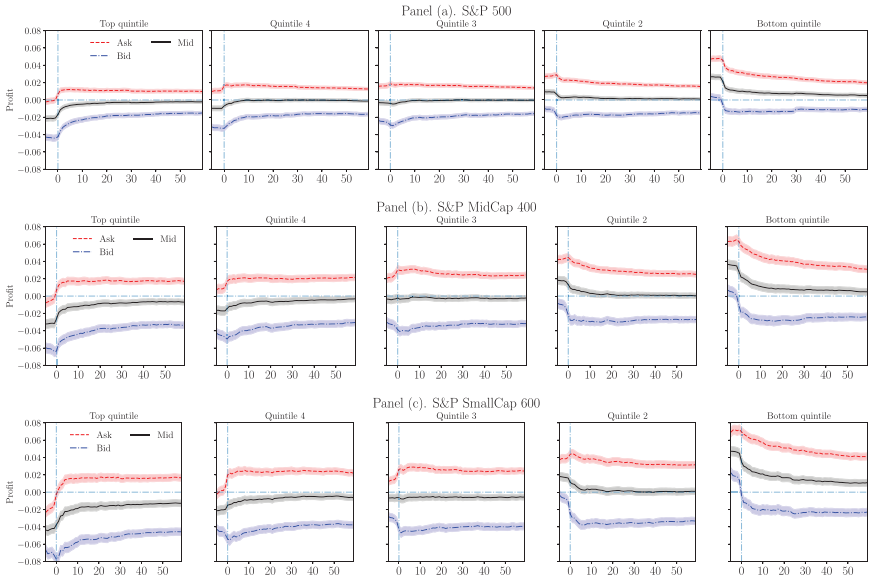

FIG 4.—Profitability around earnings announcements for displayed quotes in the after-hours market. (A) Full sample. (B) S&P 500. (C) S&P MidCap 400. (D) S&P SmallCap 600. This figure shows the average ask profit (dashed red line), bid profit (dashed-dotted blue line), and midquote profit (solid black line) and their respective 95% confidence intervals by earnings surprise quintile (columns) and by S&P index (rows) −5 to 60 minutes around earnings announcements in the after-hours market. The profits are calculated from the perspective of a seller. We define the ask profit as $(Ask\ price_{i,t} - Closing\ price_{i,T})/Ask\ price_{i,t}$, the bid profit as $(Bid\ price_{i,t} - Closing\ price_{i,T})/Bid\ price_{i,t}$, and the midquote profit as $(Midquote_{i,t} - Closing\ price_{i,T})/Midquote_{i,t}$, where $t < T$. The closing price is the official closing price of the following regular trading hour session at 4 p.m. The earnings announcement occurs at time 0. The sample period is January 1, 2011 to December 31, 2015.

4.2 HOW DO BID AND ASK PRICES ADJUST AROUND EARNINGS ANNOUNCEMENTS?

The previous result shows the price reaction in midquotes is faster than in trade prices, but that price drifts occur in midquotes following announcements, which points to slow price discovery and potential trading opportunities. However, examining changes in midquote prices around earnings announcements provides only a partial view of the price discovery process. Because trades occur at the bid or the ask price, it is essential to examine how bid and ask prices adjust to earnings surprises to have a complete understanding of price formation and its mechanism.[17] To do so, we calculate

[17] Lee, Mucklow, and Ready [1993] were one of the first to examine quoted spread around earnings announcements during regular trading hours and document large quoted spreads around announcements.

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the following measures of profitability from the perspective of a seller:

$$\begin{aligned} Ask\ profit_{i,t} &= \frac{Ask\ price_{i,t} - Closing\ price_{i,T}}{Ask\ price_{i,t}}, \\ Bid\ profit_{i,t} &= \frac{Bid\ price_{i,t} - Closing\ price_{i,T}}{Bid\ price_{i,t}}, \\ Midquote\ profit_{i,t} &= \frac{Midquote_{i,t} - Closing\ price_{i,T}}{Midquote_{i,t}}, \end{aligned} \tag{2}$$

where $t < T$. $Ask\ profit_{i,t}$ denotes the return for a liquidity provider that sells a stock at time t on earnings announcement i with a trade price occurring at the ask and "exits" its position at the closing of markets (4 p.m.) of the following regular trading hour session.[18] Similarly, $Bid\ profit_{i,t}$ denotes the return for a liquidity taker that sells a stock at time t with a trade price occurring at the bid. The $Midquote\ profit_{i,t}$ denotes the profits for a trader that sells at time t with a trade price at the midquote. The approach specified in (2) allows us to examine the adjustment in bid and ask prices and the spreads while controlling for price levels in the cross-section of stocks.

Figure 4 shows the cross-sectional average of the one-minute ask, bid, and midquote profits by S&P index and by earnings surprise quintile starting 5 minutes before to 60 minutes after earnings announcements. The red dashed line, blue dash-dotted line, and solid black line represent the ask, bid, and miquote profits, respectively, and their respective 95% confidence intervals.

Four important findings emerge from Figure 4. First, the figure depicts wide spreads around earnings announcements where the difference between the ask and bid profit lines can be as large as 6%, which suggests liquidity providers face adverse selection before the arrival of news. Spreads are wider before announcements and narrow more slowly after announcements for small firms. Second, the narrowing of spreads is asymmetric, that is, it is driven by one side of the quote. For example, following positive news, ask prices adjust almost instantaneously to the final ask price and the bid side of the quote remains to adjust. Conversely, following negative news, bid prices adjust almost instantaneously to the final bid price and the ask side of the quote remains to adjust. The gradual adjustment on the other side of the quote is what gradually pushes the midquote to the post-announcement price, generating price drifts. Third, these wide spreads suggest that, on average, it is not possible for a liquidity taker to trade profitably following earnings announcements in the after-hours market (assuming the liquidity taker exits its position at 4:00 p.m.). To see this more clearly, notice the ask (bid) profit lines are never significantly below (above) zero after the announcement. For example, for a surprise in the top quintile for an S&P 500 stock, a seller who sells at the ask (provides

[18] We obtain similar results using the opening price at 9:30 a.m. instead of the closing price.

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liquidity) during the first 60 minutes earns a profit of about 1%. A liquidity provider that buys at the bid earns 4% at the announcement and the profits decrease to 2% following the announcement; conversely, a liquidity taker that sells at the bid incurs the corresponding losses. Finally, in most cases, the post-announcement closing price (at 4 p.m.) is already within the spreads before the announcement. Therefore, on average, liquidity takers that are informed about upcoming surprises have no opportunity to trade profitably in the minutes leading to earnings announcements.[19] Section IA3 of the online appendix further shows similar quote adjustments for microcap stocks, but the adjustment in midquotes is slower because of a slower adjustment of the quote opposite to the news (i.e., bid price following positive news and ask price following negative news.)

Overall, figure 4 provides key insights into the price discovery process. In contrast to the model of Kim and Verrecchia [1994] in which trading is necessary for price discovery following earnings announcements, our findings show quote adjustments are the main drivers of price discovery, consistent with recent models of price formation (Hoffmann [2014], Budish, Cramton, and Shim [2015], Menkveld and Zoican [2017]).

We conjecture that the asymmetric price adjustment causing the slow adjustment in midquotes is most likely because only the ask following positive news or the bid following negative news is prone to sniping by fast liquidity takers if left unadjusted. Why one side of the quotes is slower to adjust can be attributed to imperfect competition or unresolved uncertainty.[20] The fact that asymmetries vary with firm size (a reasonable proxy for competition among liquidity providers or uncertainty) is suggestive evidence of imperfect competition or unresolved uncertainty. To our knowledge, we are the first to document asymmetric adjustments in *quotes* following the arrival of news.

4.3 QUANTIFYING THE DYNAMICS OF PRICE DISCOVERY

The previous section shows one side of the quote responds quickly to the news, whereas midquote and trade returns lag the best quote adjustment (the side of the quote that is in the same direction as the earnings surprise). We next quantify the speed at which trade, midquote, and best quote returns reflect earnings surprises.

We estimate the following regression:

$$r_{i,t} = \alpha + \sum_{k=1}^{30} \beta_k Surprise_i \mathbf{1}_{k=t} + \varepsilon_{i,t}, \tag{3}$$

[19] Conditioning on random-walk earnings surprises provides similar results. See figure A.1 of the Appendix.

[20] Other types of asymmetries have been documented. For example, Green, Li, and Schurhoff [2010] document prices rise faster than they fall in the OTC bond market and this asymmetry varies with competitive conditions. Johnson and So [2018] show asymmetric trading costs before earnings announcements. This asymmetry creates a predictable upward bias in prices that increases pre-announcement, and subsequently reverses.

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where $r_{i,t}$ is a two-minute stock log return calculated using trade, midquote, and best quote prices during after hours following earnings announcements i. $t = 1$ corresponds to the return starting at the closing price (at 4 p.m.) before the announcement to one minute after the announcement, and $t = 2, 3, \ldots, 30$ correspond to two-minute intervals starting one minute after the announcement and ending 59 minutes after the announcement.[21] $Surprise_i$ is the earnings surprise and $\mathbf{1}_{k=t}$ is an indicator variable equal to 1 if $k = t$ and zero otherwise. Best quote returns are calculated using ask prices for earnings surprises in the top two quintiles, midquote prices for the middle quintile, and bid prices for the bottom two quintiles. Trade returns are calculated from trade prices; for a minute interval t with no trade, we impose a return of 0%. The β_k are estimated separately for S&P 500, S&P MidCap 400, and S&P SmallCap 600 stocks.

Table 3 reports the β_k for the first eight intervals ($k = 1, 2, .., 8$) up to 15 minutes following the announcement. The table further reports the total response from the closing price before the announcement to 15 minutes ($\sum_{k=1}^{8} \beta_k$), 29 minutes ($\sum_{k=1}^{15} \beta_k$), and 59 minutes ($\sum_{k=1}^{30} \beta_k$) following the announcement. Three findings emerge from this table. First, the initial response coefficient to earnings surprises ($[Close, 1]$) for best quotes is immediate with little to no adjustment remaining in the direction of earnings surprises. Moreover, the relation between best quote returns and earning surprises turns negative following the initial response, indicating the initial response overreacts to surprises. Comparing the initial response and the total response after one hour ($\sum_{k=1}^{30} \beta_k$) suggests an overreaction of 20%, 5%, and 13% for S&P 500, 400, and 600 stocks, respectively. Such overreaction is indicative of liquidity providers setting higher ask (bid) prices following positive (negative) news to protect themselves from opportunistic traders that are more sophisticated or quicker at processing public news. Second, comparing the adjustment in midquotes versus trades shows the size of the downward bias when measuring price discovery using trades. The difference in the total adjustment between midquote and trade returns for S&P 500 stocks after one hour, $\sum_{k=1}^{30} \beta_k = 2.38$ for midquotes compared to $\sum_{k=1}^{30} \beta_k = 2.34$ for trades, is minimal; however, the difference is large for S&P MidCap 400 and 600 SmallCap stocks. The total one-hour adjustment in trade returns compared to midquote returns is 17% (1-2.05/2.46) and 35% (1-1.19/1.84) smaller in magnitude for S&P 400 and 600 stocks, respectively. Finally, the adjustments in the best quote and the midquote converge after one hour. This suggests that, immediately after the announcement, the true adjustment in fundamentals to earnings surprises is somewhere between the best quote and the midquote.

[21] We choose to start with the closing price because some stocks will have no trades or quotes between the close and the announcement.

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TABLE 3

Trade, Midquote, and Best Quote Return Response to Earnings Surprises

	S&P 500			S&P MidCap 400			S&P SmallCap 600		
Minutes	Trade (1)	Midquote (2)	Quote (3)	Trade (4)	Midquote (5)	Quote (6)	Trade (7)	Midquote (8)	Quote (9)
[*Close*,1]	1.19**	0.94**	3.25**	0.73**	0.82**	3.35**	0.38**	0.45**	2.47**
[1,3]	0.29**	0.40**	0.08	0.13*	0.36**	0.14*	0.12**	0.35**	0.10*
[3,5]	0.12**	0.23**	0.01	0.13**	0.22**	0.06	0.12**	0.26**	0.10**
[5,7]	0.07	0.13**	−0.05	0.12**	0.21**	0.07	0.04	0.13**	0.01
[7,9]	0.08*	0.04	−0.05	0.11**	0.10*	−0.04	0.04	0.11**	−0.04
[9,11]	0.02	0.10**	−0.05	0.02	0.06	−0.05	0.02	0.07	−0.06*
[11,13]	0.06*	0.07*	−0.01	0.08**	0.10**	0.01	0.03	0.05	−0.03
[13,15]	0.04	0.05	−0.06*	−0.05	0.05	−0.03	0.00	0.01	−0.02
$\sum_{k=1}^{K=8} \beta_k$	1.87	1.96	3.12	1.28	1.92	3.51	0.76	1.43	2.52
$\sum_{k=1}^{K=15} \beta_k$	2.09	2.20	2.98	1.57	2.25	3.38	0.90	1.73	2.39
$\sum_{k=1}^{K=30} \beta_k$	2.34	2.38	2.61	2.05	2.46	3.17	1.19	1.84	2.14

This table reports the estimated response coefficients β_k of the regression:

$$r_{i,t} = \alpha + \sum_{k=1}^{30} \beta_k Surprise_i \mathbf{1}_{k=t} + \varepsilon_{i,t},$$

where $r_{i,t}$ is a 2-minute stock log return calculated using trade, midquote, and best quote prices during the after-hours period following earnings announcement i. $t = 1$ corresponds to the return starting at the closing price (at 4 p.m.) before the announcement to one minute after the announcement, and $t = 2, 3, \ldots, 30$ corresponds to 2-minute intervals starting one minute after the announcement and ending 59 minutes after the announcement. $Surprise_i$ is the earnings surprise and $\mathbf{1}_{k=t}$ is an indicator variable equal to one if $k = t$ and zero otherwise. Trade returns are calculated from trade prices, and for any minute interval t with no trades, we impose a return of 0%. Midquote returns are calculated from midquote prices (i.e., the mid-point between the best displayed bid and best ask price). Best quote returns are calculated using ask (bid) prices for earnings surprises in the top (bottom) two quintiles and midquote prices for the middle quintile. $\sum_{k=1}^{8} \beta_k$, $\sum_{k=1}^{15} \beta_k$, and $\sum_{k=1}^{30} \beta_k$ correspond to the total response from the closing price before the announcement to 15, 29, and 59 minutes following the announcement, respectively. The β_k coefficients are estimated separately for S&P 500, S&P MidCap 400, and S&P SmallCap 600 stocks. ** and * indicate a two-tailed test significance level of less than 1 and 5%, respectively. Standard errors are calculated using the Driscoll and Kraay [1998] method. The sample period is January 1, 2011 to December 31, 2015.

4.4 THE ROLE OF LIQUIDITY IN PRICE DISCOVERY

Slow price discovery following news is commonly attributed to illiquidity, for example, high transaction costs (Ng, Rusticus, and Verdi [2008], Chordia et al. [2009]). Consistent with the past literature, figure 3 shows a more gradual price discovery process for smaller firms for returns computed from both midquotes and transaction prices. We next examine the role of liquidity in after-hours price discovery. Specifically, we investigate whether the bias, that is, the difference in the price formation for returns computed from quotes and trade prices, is statistically significant and whether liquidity influences the bias.

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Using firm size as a proxy for liquidity,[22] table 4 reports the coefficients of the following regression:

$$\begin{aligned} BHAR[-1, T]_{i,k} = \alpha &+ \beta_1 Surprise_i + \beta_2 Surprise_i \times MCAP_i \\ &+ \beta_3 Surprise_i \times \mathbf{1}_{[\text{Quote}]} + \beta_4 Surprise_i \times MCAP_i \\ &\times \mathbf{1}_{[\text{Quote}]} + \beta_5 MCAP_i \times \mathbf{1}_{[\text{Quote}]} + \beta_6 \mathbf{1}_{[\text{Quote}]} \\ &+ \beta_7 MCAP_i + \varepsilon_{i,k}, \end{aligned} \tag{4}$$

where $BHAR[-1, T]_{i,k}$ corresponds to buy-and-hold returns from one minute before to T minutes after the announcement i calculated using trade, midquote, or best quote denoted by subscript k. We compute $BHAR$ for T equal to 5, 20, and 60 minutes. $MCAP$ is the stock market capitalization size quintile where the highest quintile corresponds to the largest firms. In columns 1–3 of table 4, the regressions include returns k computed using trades and midquote returns. In columns 4–6, the regressions include returns computed using trades and best quote returns. $\mathbf{1}_{[\text{Quote}]}$ is a dummy equal to 1 if return k is computed using midquotes in columns 1–3 and best quotes in columns 4–6, and zero otherwise.[23]

Columns 1–3 indicates that both estimated coefficients for $Surprise_i \times MCAP_i$ and $Surprise_i \times \mathbf{1}_{[\text{Quote}]}$ are positive and statistically significant at the 1% level and that stock prices for larger (liquid) firms and returns computed using midquotes, as opposed to trade prices, reflect earnings news more quickly. For example, column 3 shows a one quintile increase in market capitalization increases the 60-minute price response to news by 21.1% (=0.485/2.297). When returns are calculated using midquotes, the response to news increases by 66% (=1.523/2.297). The estimated coefficients $Surprise_i \times MCAP_i \times \mathbf{1}_{[\text{Quote}]}$ are negative and statistically significant at the 1% level for T equal to 20 and 60 minutes (columns 2 and 3). This result has two implications. First, the relation of stock liquidity to the speed of price discovery is significantly weakened when returns are computed using midquotes as opposed to trades. Column 3 shows the effect of firm size to the response to earnings surprises decreases by 55% (=−0.267/0.485) when returns are calculated using midquotes. Second, the bias from using trade returns as opposed to midquote returns worsens for smaller and less liquid firms. For example, columns 3 reports that, for a liquid stock (top size quintile), the difference in the price adjustment from using midquote or trade prices is not statistically different from

[22] Firm size corresponds to the market capitalization a day prior to the earnings announcement. Figure IA4 in the online appendix shows the after-hours relative bid–ask spreads around earnings announcements decreases with firm size. A negative relation between transaction costs (illiquidity) and firm size is been well established in the literature (e.g., Demsetz [1968], Stoll and Whaley [1983]).

[23] Note the number of observations (N) increases in T as more stocks have after-hours observations (i.e., valid bid and ask price).

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TABLE 4

Liquidity and Price Response to Earnings Surprises

	Midquotes			Best quotes		
Quote:	5 minutes	20 minutes	60 minutes	5 minutes	20 minutes	60 minutes
T:	(1)	(2)	(3)	(4)	(5)	(6)
Surprise	1.044***	1.356***	2.297***	1.044***	1.356***	2.297***
	(0.22)	(0.24)	(0.26)	(0.22)	(0.24)	(0.26)
Surprise × MCAP	0.496***	0.604***	0.485***	0.496***	0.604***	0.485***
	(0.09)	(0.10)	(0.11)	(0.09)	(0.10)	(0.11)
Surprise $\times \mathbf{1}_{[\text{Quote}]}$	0.638***	1.306***	1.523***	4.596***	4.988***	4.753***
	(0.15)	(0.17)	(0.16)	(0.32)	(0.32)	(0.31)
Surprise × MCAP $\times \mathbf{1}_{[\text{Quote}]}$	−0.033	−0.190***	−0.267***	−0.310**	−0.547***	−0.622***
	(0.05)	(0.05)	(0.05)	(0.12)	(0.11)	(0.10)
MCAP $\times \mathbf{1}_{[\text{Quote}]}$	−0.000	0.000	0.000	−0.000	0.000	0.000
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
MCAP	0.001***	0.001**	0.001*	0.001***	0.001**	0.001*
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
$\mathbf{1}_{[\text{Quote}]}$	−0.001	−0.001	−0.001*	−0.003***	−0.004***	−0.004***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Intercept	−0.004***	−0.004***	−0.005***	−0.004***	−0.004***	−0.005***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
N	22,708	26,094	31,276	22,708	26,094	31,276
R^2	0.064	0.086	0.099	0.148	0.155	0.157

This table reports the estimated response coefficients of the regression:

$$\begin{aligned} BHAR[-1, T]_{i,k} = \alpha &+ \beta_1 Surprise_i + \beta_2 Surprise_i \times MCAP_i \\ &+ \beta_3 Surprise_i \times \mathbf{1}_{[\text{Quote}]} + \beta_4 Surprise_i \times MCAP_i \times \mathbf{1}_{[\text{Quote}]} \\ &+ \beta_5 MCAP_i \times \mathbf{1}_{[\text{Quote}]} + \beta_6 \mathbf{1}_{[\text{Quote}]} + \beta_7 MCAP_i + \varepsilon_{i,k}, \end{aligned}$$

where $BHAR[-1, T]_{i,k}$ corresponds to buy-and-hold returns from one minute before to T minutes after the announcement i calculated using trade, midquote, or best quote denoted by subscript k. We compute $BHAR$ for T equal to 5, 20, and 60 minutes. In Columns (1)-(3), the regressions include returns k computed using trades and midquote returns. In Columns (4)-(6), the regressions include returns computed using trade and best quote returns. $MCAP$ is the stock market capitalization size quintile where the highest quintile corresponds to the largest firms. $\mathbf{1}_{[\text{Quote}]}$ is a dummy equal to one if return k is computed using midquote in columns (1)-(3) and best quotes in columns (4)-(6), and zero otherwise. ** and * indicate a two-tailed test significance level of less than 1 and 5%, respectively. Standard errors are clustered by firm and earnings announcement date. The sample period is January 1, 2011 to December 31, 2015.

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zero ($1.523-(0.267\times5)=0.188$, $p = 0.102$), but the difference is larger for smaller and less liquid stocks.

Columns 4–6 report the difference in the response to news from using returns computed from best quote returns relative to trade returns. As in columns 1–3, the estimated coefficients for $Surprise_i \times MCAP_i$ and $Surprise_i \times \mathbf{1}_{[\text{Quote}]}$ are positive and statistically significant at the 1% level and indicate that stock prices for larger (liquid) firms and best quote prices, as opposed to trade prices, reflect earnings news more quickly. The estimated coefficients $Surprise_i \times \mathbf{1}_{[\text{Quote}]}$ indicate the return response to earnings surprises is approximately 2–4 times greater for best quote returns than trade returns. The estimated coefficients $Surprise_i \times MCAP_i \times \mathbf{1}_{[\text{Quote}]}$ is negative and statistically significant at the 5% and 1% levels and carries the same interpretation as in columns 1–3; that is, the bias from using trade returns as opposed to best quote returns worsens for smaller and less liquid firms.

In sum, the results reported in table 4 indicate that any variables that are correlated with size and liquidity and interact with estimates of market reactions to news will be misspecified when using trade prices instead of quotes.

4.5 ASYMMETRIC PRICE ADJUSTMENT TO POSITIVE AND NEGATIVE NEWS

The literature documents asymmetric price reaction to good and bad earnings news. For example, Kothari, Shu, and Wysocki [2009] document greater price reactions following negative news than positive news because managers delay disclosure of bad news relative to good news on days leading to earnings announcements. Johnson and So [2018] find the cost of trading on bad news, relative to good news, increases before earnings announcements. The authors argue this asymmetry is due to liquidity providers reducing their exposure to earnings announcement risk by providing liquidity asymmetrically. Our analysis of price formation in the after-hours market allows us to investigate whether liquidity providers are more averse to providing liquidity following bad earnings news than good news. Following earnings announcements, other news is anticipated to be released, for example, conference calls and analyst recommendations. We thus expect liquidity providers to be more averse to trading against sellers and increase their inventory following negative news. Consequently, as in Johnson and So [2018], liquidity providers will charge a relatively higher trading price after negative news than positive news.

To examine the asymmetric response to earnings news, we follow Andersen et al. [2003] and examine the impact of stock return responses to earnings surprises, allowing for a different constant and slope for good and bad earnings news. The literature clearly indicates analyst short-term forecasts tend to be pessimistic (e.g., Matsumoto [2002], Chan, Karceski, and Lakonishok [2007]), and firms that just meet earnings expectations generate negative returns. Therefore, we examine the response to positive and negative

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normalized earnings surprises after adjusting for potential pessimism in analyst forecasts. The normalized earnings surprise is defined as

$$\langle Surprise \rangle_i = \frac{Surprise_i - Surprise^*}{\sigma(Surprise)}, \tag{5}$$

where σ is the standard deviation and $Surprise^*$ corresponds to the earnings surprise for which $E[Return|Surprise^*] = 0$. We estimate $Surprise^*$ by first regressing after-hours returns calculated using the closing price (4 p.m.) to the next day opening price (9:30 a.m.) around earnings announcements on earnings surprises estimated for the full sample and by S&P indices. We then solve for the zero-return earnings surprise ($Surprise^*$) using the parameter estimates.[24] We then estimate the following earnings response specification:

$$BHAR[-1, 60]_i = \begin{cases} \beta_1 \langle Surprise \rangle_i + \beta_2 \langle Surprise \rangle_i^2 + \varepsilon_i, & \text{if } \langle Surprise \rangle_i \leq 0 \\ \beta_3 \langle Surprise \rangle_i + \beta_4 \langle Surprise \rangle_i^2 + \varepsilon_i, & \text{if } \langle Surprise \rangle_i > 0, \end{cases} \tag{6}$$

where $BHAR[-1, 60]$ corresponds to the buy-and-hold return 1 minute before to 60 minutes following an earnings announcement using trade, midquote, or best quote returns.

We graphically represent the regression estimates $\beta_1 \langle Surprise \rangle_i + \beta_2 \langle Surprise \rangle_i^2$ to the left of the origin and $\beta_3 \langle Surprise \rangle_i + \beta_4 \langle Surprise \rangle_i^2$ to the right of the origin in figure 5 as "earnings news impact curves" for the full sample and by S&P index.

Three main findings stand out. First, the figure shows the effect of earnings news is asymmetric: negative surprises have greater impact than positive surprises and the asymmetry is more pronounced when returns are computed using best quotes. This suggests liquidity providers charge a higher price (i.e., lower bid price to sellers) following negative news. Second, the response to positive and negative news is more asymmetric for smaller firms. Finally, the difference between the return responses computed from midquotes and best quotes for positive news is minimal but for negative news is substantial. To summarize, these results indicate liquidity providers provide liquidity asymmetrically not only leading to announcements as in Johnson and So [2018] but also following announcements to minimize the risk associated with upcoming news (e.g., conference calls and analyst recommendations).

4.6 How profitable are liquidity providers' executed orders?

Our previous results suggest liquidity providers are sophisticated at processing and adjusting *displayed* quotes to earnings surprises. Therefore, we should expect their executed orders are profitable on average because of

[24] The specific values for $Surprise^*$ for the full sample, S&P 500, S&P 400, and S&P 600 index are 0.00089, 0.00079, 0.00094, and 0.00092, respectively.

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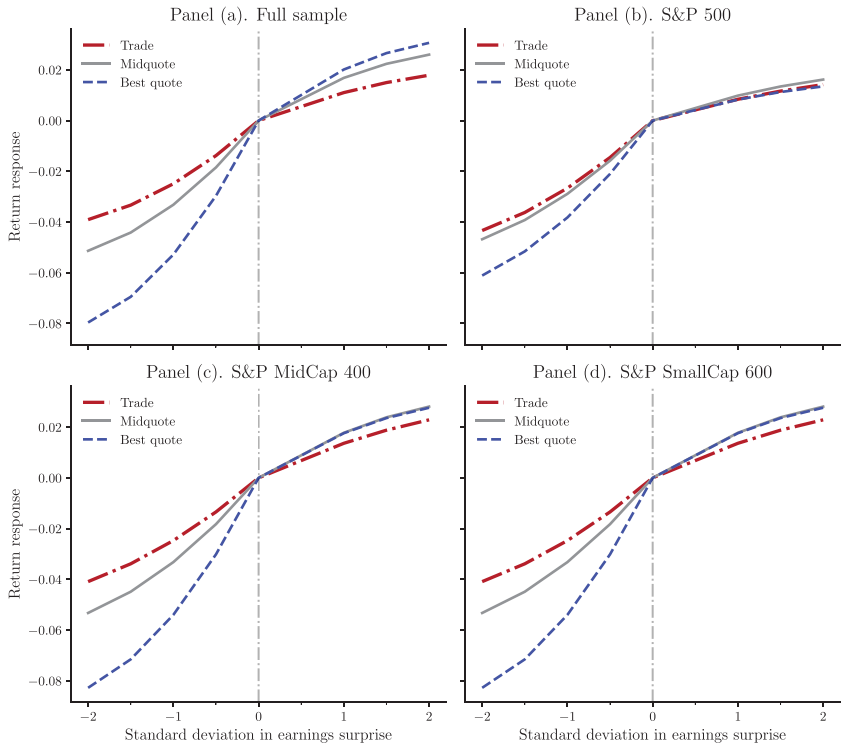

FIG 5.—The earnings news response curve. (A) Full sample. (B) S&P 500. (C) S&P MidCap 400. (D) S&P SmallCap 600.
This figure shows the earnings news reponse curve: $\beta_1\langle Surprise\rangle_i + \beta_2\langle Surprise\rangle_i^2$ to the left of the origin and $\beta_3\langle Surprise\rangle_i + \beta_4\langle Surprise\rangle_i^2$ to the right of the origin estimated from the following earnings response specification:

$$BHAR[-1,60]_i = \begin{cases} \beta_1\langle Surprise\rangle_i + \beta_2\langle Surprise\rangle_i^2 + \varepsilon_i, & \text{if } \langle Surprise\rangle_i \leq 0 \\ \beta_3\langle Surprise\rangle_i + \beta_4\langle Surprise\rangle_i^2 + \varepsilon_i, & \text{if } \langle Surprise\rangle_i > 0. \end{cases}$$

$BHAR[-1,60]$ corresponds to the buy-and-hold return 1 minute before to 60 minutes following an earnings announcement using trade, midquote, or best quote returns. $\langle Surprise\rangle$ corresponds to the normalized earnings surprise defined in equation (5). The results are presented for the full-sample, S&P 500, S&P 400, and S&P 600 stocks in panels A to D, respectively. The sample period is January 1, 2011 to December 31, 2015.

the large spreads and fast adjustments in quotes following announcements. However, the use of *hidden* limit orders is prevalent. We find about 40% of executed limit orders on NASDAQ following earnings announcements in the after-hours market are hidden, while 25% of trades execute against hidden orders when no earnings announcements occur, and the fraction is only 12% during regular trading hours. Chakrabarty and Shaw [2008] also find the use of hidden orders increases on earnings announcement days.

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For a hidden order to be executed, it must be placed within the spread because displayed quotes always have priority at the same price. As such, hidden orders could be placed by unsophisticated liquidity providers and executed at a loss, and therefore trades contribute to price discovery when executed against hidden orders. However, the literature often associates hidden orders with sophisticated traders Bloomfield, O'Hara, and Saar [2015].[25] Examining the profitability provides a robustness check for the interpretation of our previous results, that is, that liquidity providers are sophisticated at processing news.

To measure the profitability of liquidity providers associated with displayed and hidden orders, we calculate, for each observed after-hours trade j following each earnings announcement i, the realized spread measure $rs_{i,j}$ defined as

$$rs_{i,j} = \begin{cases} (m_i - p_{i,j})/p_{i,j} \times 100, & \text{if trade } j \text{ is a passive buy,} \\ (p_{i,j} - m_i)/p_{i,j} \times 100, & \text{if trade } j \text{ is a passive sell,} \end{cases} \tag{7}$$

where m_i is the crossing price at the opening of markets following the earnings announcement if there is an auction or midquote in the order book at 9:30 a.m. otherwise and $p_{i,j}$ is the trade price. We further winsorize the realized spreads at the 1st and 99th percentiles. We examine the profitability of liquidity providers for two time intervals: the first five minutes and from the 5th to the 60th minute following earnings announcements. For each of the time intervals following earnings announcement i, we compute the average realized spread, $\overline{rs}_{i,k}$, across trades against limit orders of type $k \in$\{all order types, hidden, displayed, displayed stale, and nonstale displayed orders\} by S&P index. We define stale orders as limits orders posted before the announcement. Baldauf and Mollner [2021] show, in the context of a macroeconomic news, that stale orders are quickly snipped by high-frequency traders following the arrival of news (i.e., the liquidity provider is incurring a loss).

Table 5 reports the mean $\overline{rs}_{i,k}$ across all order types in panel A, for hidden and displayed orders in panel B, and for displayed stale and nonstale orders in panel C. Columns 1, 3, and 5 of panel A report that, in the first five minutes, liquidity providers earn profits of approximately −5, −24, and −87 bps on average for S&P 500, S&P MidCap 400, and S&P SmallCap 600 stocks, respectively, but only the profits for S&P 400 and 600 stocks are statistically significant at the 5% and 1% levels, respectively. Columns 2, 4, and 6 present statistically significant positive profits from the 5th to the 60th

[25] Bloomfield, O'Hara, and Saar [2015] show in a lab experiment that informed traders prefer hidden orders so as not to reveal how much they are willing to buy or sell and earn higher profits. Harris [1996] and Bessembinder, Panayides, and Venkataraman [2009] argue hidden orders are effective for mitigating adverse selection. Boulatov and George [2013] develop a model and show the presence of hidden liquidity intensifies competition among informed traders and improves market efficiency. Yao [2017] also shows empirically that informed traders use hidden orders.

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TABLE 5

Realized Spread by Order Type

Panel A: All Order Types

	S&P 500		S&P MidCap 400		S&P SmallCap 600	
	5 minutes (1)	5–60 minutes (2)	5 minutes (3)	5–60 minutes (4)	5 minutes (5)	5–60 minutes (6)
All order types	−0.051	0.074**	−0.241*	0.182**	−0.866**	0.170*
N	2,871	4,600	1,174	1,936	1,483	2,348

Panel B: Hidden and Displayed Orders

	S&P 500		S&P MidCap 400		S&P SmallCap 600	
	5 minutes (1)	5–60 minutes (2)	5 minutes (3)	5–60 minutes (4)	5 minutes (5)	5–60 minutes (6)
Hidden orders	0.100*	0.159**	0.062	0.245**	−0.156	0.609**
Displayed orders	−0.038	0.065*	−0.331**	0.182**	−0.666**	0.159*
N hidden	1,625	2,665	577	1,049	543	1,119
N displayed	2,033	3,241	848	1,429	992	1,638

Panel C: Displayed Stale and Non-Stale Orders

	S&P 500		S&P MidCap 400		S&P SmallCap 600	
	5 minutes (1)	5–60 minutes (2)	5 minutes (3)	5–60 minutes (4)	5 minutes (5)	5–60 minutes (6)
Stale orders	−0.202*	0.049	−1.174**	−0.009	−1.215**	−0.183
Nonstale orders	0.019	0.058*	−0.161	0.146*	−0.203	0.197*
N stale	1,436	659	519	240	651	314
N nonstale	1,801	3,206	670	1,387	657	1,562

This table reports the mean realized spread across all trades against displayed, hidden, displayed non-stale, and displayed stale limit orders. We calculate the realized spread, $rs_{i,j}$, for each observed after-hours trade j following each earnings announcement i, associated with different limit order types as

$$rs_{i,j} = \begin{cases} (m_i - p_{i,j})/p_{i,j} \times 100, & \text{if trade } j \text{ is a passive buy,} \\ (p_{i,j} - m_i)/p_{i,j} \times 100, & \text{if trade } j \text{ is a passive sell,} \end{cases}$$

where m_i is the crossing price at the opening of markets following the earnings announcement if there is an auction or midquote in the order book at 9:30 a.m. otherwise and $p_{i,j}$ is the trade price. For different time intervals following earnings announcement i, we compute the average realized spread, $\overline{rs}_{i,k}$, across trades against limit orders of type $k \in$ {all order types, hidden, displayed, displayed stale, and non-stale displayed orders} by S&P index. Panel A reports the mean $\overline{rs}_{i,k}$ across all order types, Panel B reports the mean $\overline{rs}_{i,k}$ for hidden and displayed orders, and Panel C reports the mean $\overline{rs}_{i,k}$ for displayed stale and non-stale orders for the first 5 minutes and from 5 to 60 minutes following earnings announcements. ** and * indicate a two-tailed test significance level of less than 1 and 5%, respectively, using standard errors clustered by date estimated by pooled regression. We also report the total number of $\overline{rs}_{i,k}$ observations (N) corresponding to each order type k. The sample period is January 1, 2011 to July 13, 2014.

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minute following earnings announcements, varying from 7 to 18 bps across all S&P indexes. At first glance, these results suggest liquidity providers are operating at a loss against sophisticated liquidity takers at the time of the announcement. However, these results ignore the heterogeneity in sophistication among liquidity providers. When examining the profitability associated with hidden and displayed orders in panel B, we find liquidity providers earn statistically significant profits on hidden orders in the first five minutes for S&P 500 stocks of 10 bps; in the following 55 minutes, they earn profits varying from 16 to 61 bps across all S&P indexes. Nonetheless, displayed orders for S&P 400 and S&P 600 stocks remain associated with negative profits in the first five minutes following announcements. Differentiating between stale and nonstale displayed orders is important. Panel C reports that, in the first five minutes following the announcement, liquidity providers' profits for nonstale orders are not statistically different from zero while profits for stale displayed orders are negative and statistically significant, varying from −20 to −122 bps. In the following 55 minutes, panel C shows displayed nonstale orders are profitable, with profits varying from 6 to 20 bps, whereas stale orders are not statistically different from zero.

Overall, our results on profitability present additional evidence that liquidity providers following earnings announcements are generally sophisticated. The fact that hidden orders are associated with positive returns confirms previous theoretical and empirical research that suggests informed traders use hidden orders. These results beg the question: Why is not there more entry by liquidity providers to compete away these profits? A possible explanation is the large fixed cost to enter the realm of high-frequency liquidity provision, for example, investment in low-latency technology. Moreover, because earnings announcements are followed by more news (e.g., conference calls), liquidity providers face inventory risk and thus must be compensated for this risk. This is an interesting area for future research.

5. *Price Formation Around Analyst Recommendations*

We next illustrate that examining quote adjustments around after-hours news is especially important when multiple confounding news events occur in the same trading session. To do so, we examine price formation around analyst recommendation revisions, which are often triggered by corporate announcements such as earnings releases (Altınkılıç and Hansen [2009], Li et al. [2015], Crane and Crotty [2020]).

Li et al. [2015] were the first to examine price formation around after-hours analyst recommendations revisions and find that after-hours revisions are associated with measurably greater overall price adjustments than regular-hour revisions. They suggest after-hours revisions have different information characteristics from regular-hour revisions and that analysts have incentives to issue revisions after hours to benefit large institutional clients. We conjecture that the use of trades to measure price reactions can be

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responsible for the significant difference in price formation for regular-hour and after-hours recommendation revisions. Our analysis is not a replication of Li et al. [2015], as we have a different time-series of analyst recommendations (our 2011–2015 vs. their 2003–2010), and we conduct our analysis in a period during which high-frequency trading is more prevalent. Furthermore, we do not condition on the direction of the news that precedes analyst revisions as they do. Our goal is to highlight that using trade returns or midquote returns can yield different inference about price formation around analyst recommendations.

We retrieve analyst recommendation revisions for our sample of S&P 1500 stocks for the period of 2011–2015 from RavenPack. As in Li et al. [2015], we select news for which a change in revision occurs, that is, a revision.[26] We retrieve a total of 19,720 analyst revisions. The fraction of revisions occurring during after hours, regular trading hours, and outside trading hours (8 p.m. to 4 a.m.) is 71.0%, 24.8%, and 4.2%, respectively.[27] Figure A.1 of the Appendix plots the total number of revisions from our sample by time of day. The majority of after-hours revisions are announced in the hours before the opening of markets, while regular-hour revisions largely occur just after the opening of markets. Because many of the recommendations are issued following firm news events (e.g., earnings announcements) that generally occur after hours, it is natural that analyst revisions are released around the opening of markets.

We begin by examining cumulative returns using trade prices around positive and negative analyst revisions over a 400-minute window before and after the event. As in Li et al. [2015], the returns at time t are computed using the most recent trade up to time t, even if that trade occurred many minutes before t. If no trade occurs in a given time interval, we impose 0%. Figure 6 shows the average cumulative returns around regular-hour and after-hours analyst revisions in panels A and B, respectively. We corroborate the findings of Li et al. [2015] and show significant price reaction and price drifts following after-hours revisions and no reaction for regular-hour revisions.

We next show in figure 7 the dynamics in quote adjustments around regular-hour (panel A) and after-hours (panel B) positive and negative analyst revisions using measures of profitability from the perspective of a seller as specified in equation (2) over a 400-minute window. We also calculate profits using trade prices. If no trades occur in a given time interval, we use the previous trade price (equivalent to assuming a return of 0%). The gray shaded area shows the fraction of observations at minute t that are during regular hour. Note that the widening of spreads, that is, the difference between the bid and ask price adjustments, depends on what fraction

[26] RavenPack indicates which news consisted of a recommendation revision and has an accurate timestamps.

[27] In Li et al. [2015], 70% of the analyst recommendations occurred after hours.

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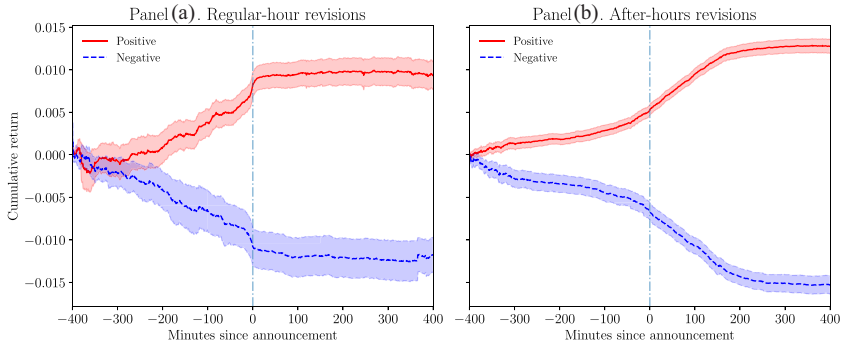

FIG 6.—Cumulative returns using trade prices around analyst recommendation revisions. (A) Regular-hour revisions. (B) After-hour revisions.
This figure shows the cumulative returns for positive (red solid line) and negative (blue dashed line) analyst recommendation revisions using trade prices 400 minutes before and following regular-hour and after-hours revisions in panels a and b, respectively. For any minute interval t with no trades, we impose a return of 0%. We only consider analyst recommendations that correspond to a change in recommendation. The sample period is January 1, 2011 to December 31, 2015.

of observations are during after hours versus regular hours. As shown in Barclay and Hendershott [2004], spreads are wide outside of trading hours and gradually narrow when approaching 9:30 a.m.

The main finding in figure 7 is that using trade prices to calculate returns biases (delays) the actual adjustment of prices around analyst revisions. Note that the dynamics in the trading profit (dotted purple line) around revisions closely map the cumulative returns using trade prices presented in figure 6. However, the dynamics in the midquote profit line present a different story of the price formation process around analyst revisions. First, as opposed to the slow price formation following after-hours revisions shown in panel B of figure 6, no slow adjustments in midquotes occur following revisions, consistent with the price adjustment observed following regular-hour revisions (see panel A of figure 6). The midquote profit is about 0% in the minutes following revisions, indicating the midquote at the time of the revision is on average equal to the final closing price at 4 p.m. Second, the difference between the midquote and trade price adjustments is wide before after-hours revisions and generally not statistically different for regular-hour revisions. The greater the difference, the more significant is the bias in price adjustments when using trade prices. The distance between the trade and midquote profit lines is larger when a greater fraction of observations (the nonshaded area) are within after hours.

Figure 6 confirms the use of trade prices to calculate returns leads to incomplete inferences regarding the price formation process following after-hours revisions. Because little to no trading often occurs, using trade returns underestimates the speed of price adjustments around after-hours revisions. If we limit ourselves to revisions issued between 5 p.m. and 8:30 a.m.

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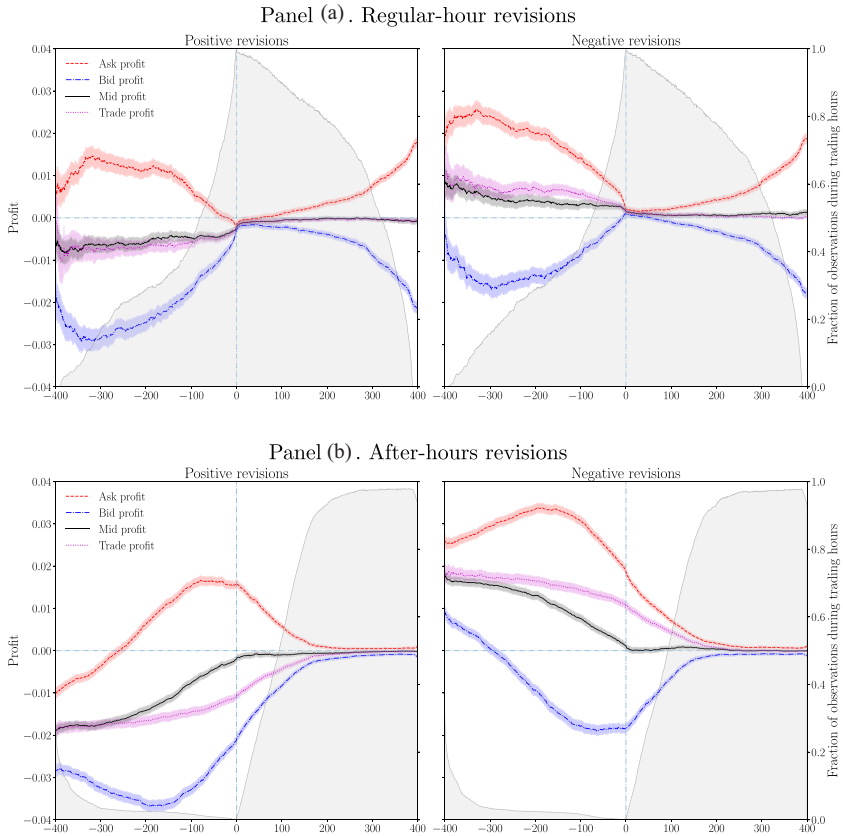

FIG 7.—Profitability around analyst recommendation revisions. (A) Regular-hour revisions. (B) After-hour revisions.

This figure shows the average ask profit (dashed red line), bid profit (dashed-dotted blue line), midquote profit (solid black line), trade profit (dotted purple line), and their respective 95% confidence intervals 400 minutes before and after positive and negative analyst recommendation revisions that occurred during regular hours (panel A) and after hours (panel B). We defined the profits from the perspective of a seller and calculate ask profit as $(Ask\ price_{i,t} - Closing\ price_{i,T})/Ask\ price_{i,t}$, the bid profit as $(Bid\ price_{i,t} - Closing\ price_{i,T})/Bid\ price_{i,t}$, the midquote profit as $(Midquote_{i,t} - Closing\ price_{i,T})/Midquote_{i,t}$, and the trade profit as $(Trade\ price_{i,t} - Closing\ price_{i,T})/Trade\ price_{i,t}$, where t defines a one-minute interval and $t < T$. The closing price is the official closing price of the regular-hour session at 4 p.m. following the revision. The gray shaded area represents the fraction of observations at minute t that occur during regular trading hours. The sample period is January 1, 2011 to December 31, 2015.

(10,289 events) and examine a 60-minute window around revisions, 44% of these events have no trade; among those events with at least one trade, the median number of trades is 9. The price reactions obtained using trade returns following analyst revisions are thus most likely due to a lagged reaction associated with the event preceding analyst revisions. Therefore, re-

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searchers should use quotes and not trade prices for a more complete inference of price formation around analyst recommendation revisions.[28]

6. *Concluding Remarks and Recommendations*

There is a growing interest in accounting research to examine the impact of corporate news events on prices using intraday data. In contrast to the seminal model of Kim and Verrecchia [1994] in which price discovery occurs through trading, we find prices adjust to earnings announcements mostly through changes in quotes by sophisticated liquidity providers. Our findings support recent models of price formation in the presence of high-frequency traders (e.g., Hoffmann [2014], Menkveld and Zoican [2017], Budish, Cramton, and Shim [2015]). We further show computing returns using trade prices as opposed to quotes biases the speed and magnitude of price discovery downward because trading is not necessary for price discovery. Such biases worsen when trading is sparse (e.g., during after-hours market, illiquid stocks). We emphasize our results do not speak to price formation studies using daily returns, for example, from CRSP. Daily returns are computed from daily closing prices, which are determined by the daily closing auction. The closing auction is usually the period of the day with the most considerable fraction of trade volume.

We propose the following guidelines for future studies examining intraday price formation:

1) *Use midquotes and not trade prices to compute stock returns.* Informed trading is endogenous: a trade generally occurs if a trader detects a profitable trading opportunity, that is, a mispriced quote. If quotes encompass the efficient price, investors have no incentive to trade, and thus measuring price discovery using trade prices can induce a bias. Because a trade is more likely to be observed when quotes are mispriced, the absence of trades can mean quotes have properly adjusted and price discovery is complete. The use of trade prices to compute returns when trading is sparse causes price adjustments to be measured with a delay compared to using quote prices. This delay can impact inference when it is sufficiently large to confound multiple news events, as we show with analyst recommendations. One should follow this recommendation, whether a news event occurs during regular hours or after hours. However, we note midquotes can take up to one hour to fully reflect news for smaller stocks in the after-hours market. This leads to our second recommendation:

[28] The use of trade prices for news events that occur during the day is not as problematic (as shown in figure 6, panel A) because the market is liquid and many more trades occur. We still, however, recommend researchers use midquotes to avoid biases related to the bid-ask bounce when using trade prices (Roll [1984]).

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2) *Use best quotes and not midquotes to compute returns around after-hours news events at very short horizons (less than one hour).* We find "best quotes" (i.e., ask for positive news and bid for negative news) react more quickly than the opposite quote, causing a delay in the adjustment of midquote prices. Therefore, it is best to use only quotes on the fast side when studying the price response for very short horizons of less than one hour. This approach has two main drawbacks. First, researchers have to *sign* the news, which is not always a trivial task. Second, because spreads decrease over time following the news, a mechanical reversal effect in best quote returns occurs for longer horizons. For these reasons, we recommend the use of midquotes for horizons of more than one hour.
3) *Examine both bid and ask quotes when studying trading opportunities around news events.* Because "best quotes" adjust quickly but the other side does not, resulting in midquote price drifts, such drifts do not imply profitable trading opportunities.
4) *After-hours news events with little to no trading should not be excluded.* Because price discovery predominantly occurs through changes in quotes, price discovery still occurs even when little to no trading occurs. Excluding those events leads to a selection bias toward larger, more liquid stocks, and events with larger news shocks.

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APPENDIX A

Low-frequency stock data sources

CRSP	Compustat
- Daily opening and closing prices	- Market capitalization - S&P index membership

Earnings and analysts data sources

I/B/E/S	RavenPack
- Earnings announcements - Analyst forecasts	- The earnings announcement millisecond timestamps - Analyst recommendation revisions with accurate timestamps

High-frequency stock data sources

Thomson Tick Reuters History (TRTH)	Nasdaq Total-View ITCH

- Quotes: bid and ask prices.
  - Only top of book (best quotes).
  - Data for all U.S. exchanges (lit venues)
- Trades: prices, volume and exchange.
  - Trades are not signed.
  - Trades against hidden orders are not flagged.
  - Data for all U.S. equity trades (inc. dark pools)
- Millisecond timestamps
	- Signed trade directions.
	- Full order book activity.
	- Flag for trades against hidden orders.
	  - Trade direction available until July 2014.
	- Nanosecond timestamps
	- Activity on the Nasdaq exchange

FIG A1.—Data sources.
This figure shows the data sources used in this study and their respective attributes exploited in this paper.

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